S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 10

Aim: Apply prompt engineering to construct a prompt for learning the concept of Adversarial Search.

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AIM: To design and refine an effective prompt using prompt engineering techniques for learning the concept of Adversarial Search from Artificial Intelligence.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- To be able to understand the concept of a *prompt* (input given to an AI model).
- To be able to understand the concept of *prompt engineering* (designing inputs to get desired outputs)
- To be able to work with basic idea and models like ChatGPT.
- To be able to distinguish between a vague prompt and a well-structured one.

THEORY:

i. Fundamentals of Prompt Engineering

Prompt engineering is the process of designing and structuring inputs (prompts) to get accurate, relevant, and well-formatted outputs from AI systems such as ChatGPT. Prompt engineering is how you ask a question to AI so that it gives the best possible answer.

ii. Characteristics of a Good Prompt

A good prompt is based on the following core principles:

- **Clarity** → Avoid ambiguity
- **Specificity** → Define exact task
- **Context** → Provide background
- **Constraints** → Limit output (length, format, style)
- **Intent** → Clearly state what is required

Example:

- Weak: *"Explain AI"*
- Strong: *"Explain AI in 5 bullet points with real-world examples for beginners"*

iii. Prompt Patterns / Techniques

The common patterns of the prompt along with their usage are as follows:

a) Role-Based Prompting

"You are an AI professor…"

b) Step-by-Step Prompting

"Explain step-by-step…"

c) Few-shot Prompting

Provide examples in the prompt

d) Instruction + Constraints

"Explain in 100 words using simple language"

e) Chain-of-Thought (Guided Reasoning)

"Show reasoning before final answer"

iv. Task-Oriented Prompting

Based on the tasks we can have different ways to provide the prompt. Following are the examples to design prompts for different tasks:

- **Learning/Explanation**
- **Coding**
- **Summarization**
- **Question generation**
- **Problem solving (AI, Cloud, etc.)**

v. Iterative Refinement (Very Important)

The first prompt may not be the best one. So, we must learn how the prompt can be refined to get most appropriate outcome:

- First prompt ≠ best prompt
- Improve by:
    - Adding constraints
    - Asking follow-up questions
    - Correcting output

Concept:

Prompt → Output → Refine → Better Output

vi. Evaluation of AI Output

It is very important to evaluate the output generated by AI. Critical thinking should be done to find the answer to the following:

- Is the answer **correct**?
- Is it **complete**?
- Is it **biased or vague**?
- Does it follow instructions?

vii. Ethical and Responsible Use

- Avoid Misuse and Plagiarism
    - Do not copy AI-generated content blindly for assignments or research
    - Always ensure originality and proper attribution when required
- Verify Information
    - AI tools like ChatGPT may generate incorrect or outdated information
    - Cross-check important facts before using them in academic or professional work
- Use AI as a Support Tool
    - AI should assist learning, not replace critical thinking
    - Students should understand concepts rather than depend entirely on AI-generated answers

viii. Basic Applications in Academics

- Generate notes
- Create MCQs and scenario-based questions
- Debug code
- Explain complex topics simply

ix. Perform the following tasks:

- Convert a bad prompt → good prompt
- Generate:
    - Notes
    - Quiz questions
    - Code snippets
- Design a prompt for:
    - Adversarial Search explanation

EXPECTED OUTPUT:

- Definition
- Game tree explanation
- Minimax working
- Alpha-Beta optimization
- Example
- Complexity analysis

INPUT & OUTPUT:

Attempt	Prompt Quality	Output Quality	Improvements
1(Basic)	"Explain Poker in AI."	Too vague, lacks structure, missing algorithms	Add specific topics (minimax, example, complexity)
2(Improved)	"Explain Poker as an adversarial search problem. Include game tree, minimax, and example."	Better, but lacks clarity on uncertainty	Add constraints: diagrams, table, structured headings
3(Refined)	"Explain Poker as an adversarial search problem in AI. Include definition, game tree, minimax, alpha-beta limitations, example with actions, and complexity analysis in structured format with tables."	High-quality, detailed, well-structured answer	Final optimized prompt

CONCLUSION:

DISCUSSION QUESTIONS:

1. What is prompt engineering?

2. Why is prompt refinement important?

3. What is Alpha-Beta pruning?

4. Difference between weak and strong prompts?

REFERENCES:

- https://learnpromptengineering.dev
- https://www.linkedin.com/pulse/40-prompt-engineering-resources-upskilling-spencer-taylor-xje0e/
- https://justprompting.in/